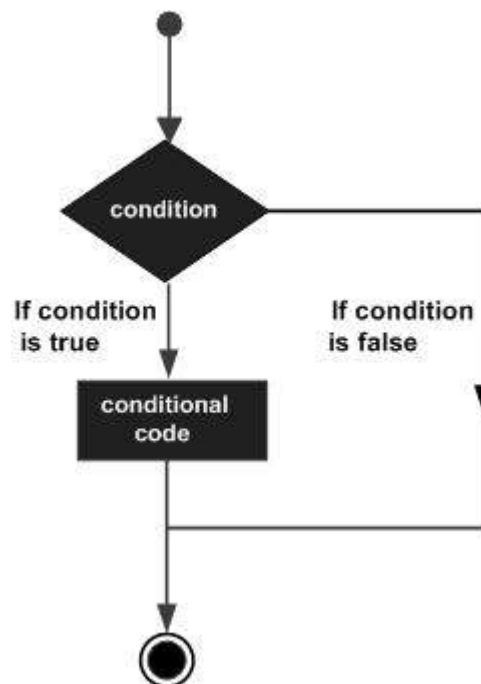


Python - Decision Making

Python's decision making functionality is in its keywords – **if..elif...else**. The **if** keyword requires a **boolean expression**, followed by colon (:) symbol. The colon (:) symbol starts an indented block. The statements with the same level of indentation are executed if the boolean expression in **if statement** is **True**. If the expression is not True (False), the **interpreter** bypasses the indented block and proceeds to execute statements at earlier indentation level.

Decision structures evaluate multiple expressions which produce TRUE or FALSE as outcome. You need to determine which action to take and which statements to execute if outcome is TRUE or FALSE otherwise.

Following is the general form of a typical decision making structure found in most of the programming languages –



Python programming language assumes any **non-zero** and **non-null** values as TRUE, and if it is either **zero** or **null**, then it is assumed as FALSE value.

Types of Decision Making Statements in Python

Python programming language provides following types of decision making statements. Click the following links to check their detail.

Sr.No.	Statement & Description
1	if statements An if statement consists of a boolean expression followed by one or more statements.
2	if...else statements An if statement can be followed by an optional else statement , which executes when the boolean expression is FALSE.
3	nested if statements You can use one if or else if statement inside another if or else if statement(s).

Let us go through each decision making briefly –

Single Statement Suites

If the suite of an **if** clause consists only of a single line, it may go on the same line as the header statement.

Example

Here is an example of a **one-line if** clause –

```
var = 100
if ( var == 100 ) : print ("Value of expression is 100")
print ("Good bye!")
```

When the above code is executed, it produces the following result –

```
Value of expression is 100
Good bye!
```

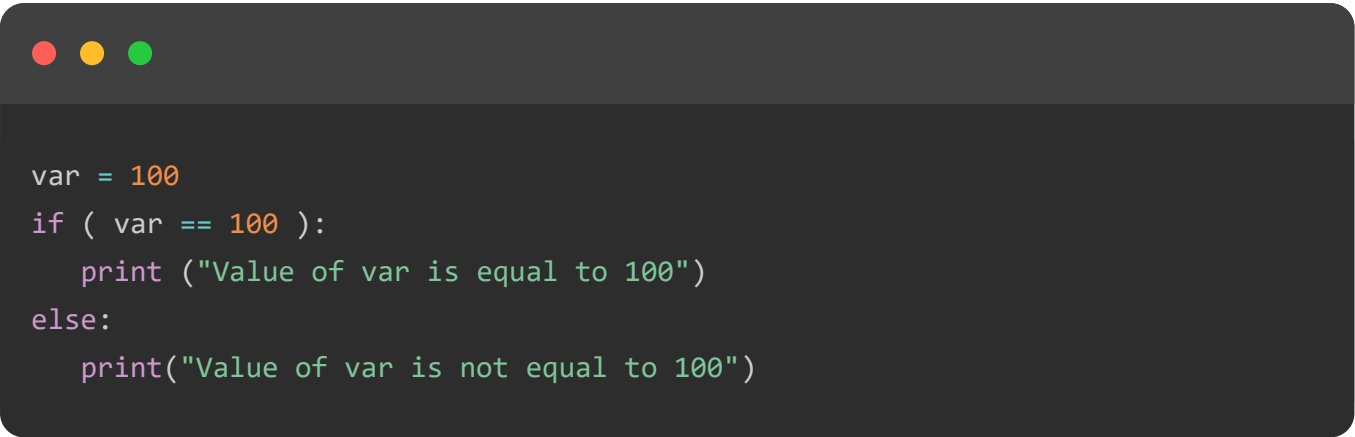
if...else statement

In this decision making statement, if the **if** condition is true, then the statements within this block are executed, otherwise, the **else** block is executed.

The program will choose which block of code to execute based on whether the condition in the **if** statement is true or false.

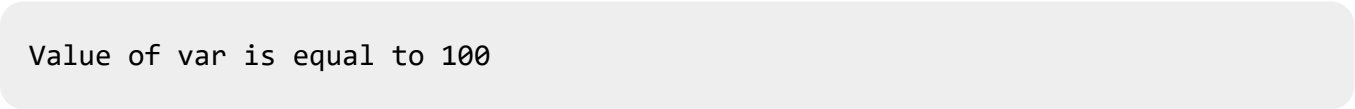
Example

The following example shows the use of if...else statement.



```
var = 100
if ( var == 100 ):
    print ("Value of var is equal to 100")
else:
    print("Value of var is not equal to 100")
```

On running the above code, it will show the following output –



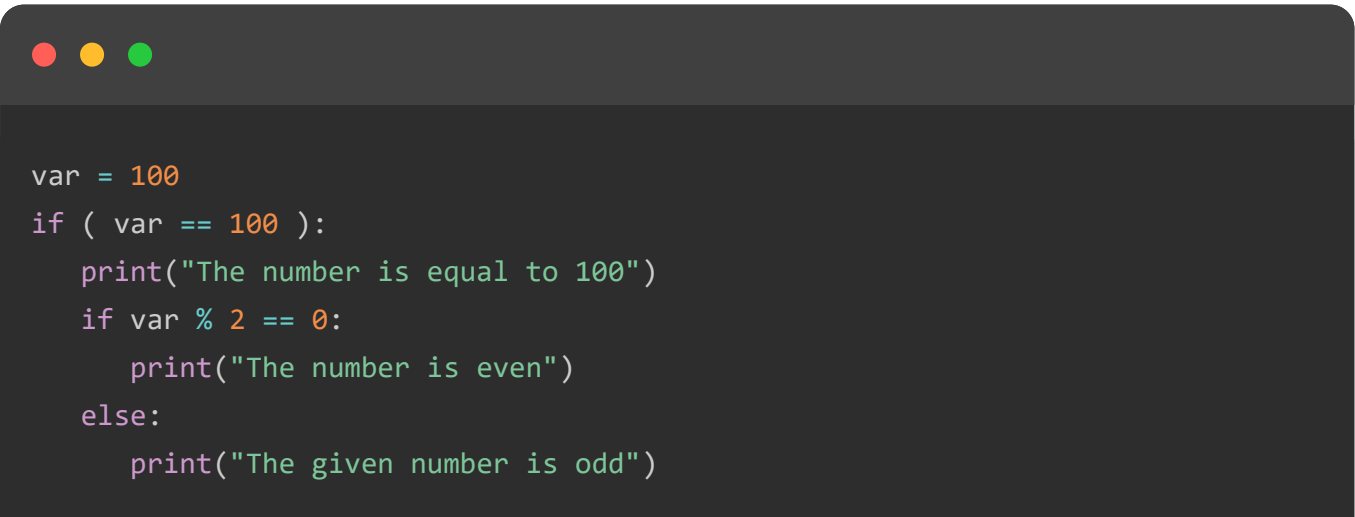
```
Value of var is equal to 100
```

Nested if statements

A nested **if** is another decision making statement in which one **if** statement resides inside another. It allows us to check multiple conditions sequentially.

Example

In this example, we will see the use of nested-if statement.



```
var = 100
if ( var == 100 ):
    print("The number is equal to 100")
    if var % 2 == 0:
        print("The number is even")
    else:
        print("The given number is odd")
```

```
elif var == 0:  
    print("The given number is zero")  
else:  
    print("The given number is negative")
```

On executing the above code, it will display the below output –

```
The number is equal to 100  
The number is even
```

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